

MICROWAVE BIPOLAR JUNCTION TRANSISTOR (BJT)

Declaration

We declare that this report is the result of our own study and research, and it has not been submitted elsewhere for academic credit. All sources of information have been duly acknowledged.

Acknowledgement

We would like to express our sincere gratitude to our instructor for the guidance and support provided during the preparation of this report. Special thanks also go to our peers for their insightful contributions and collaboration.

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1. Introduction

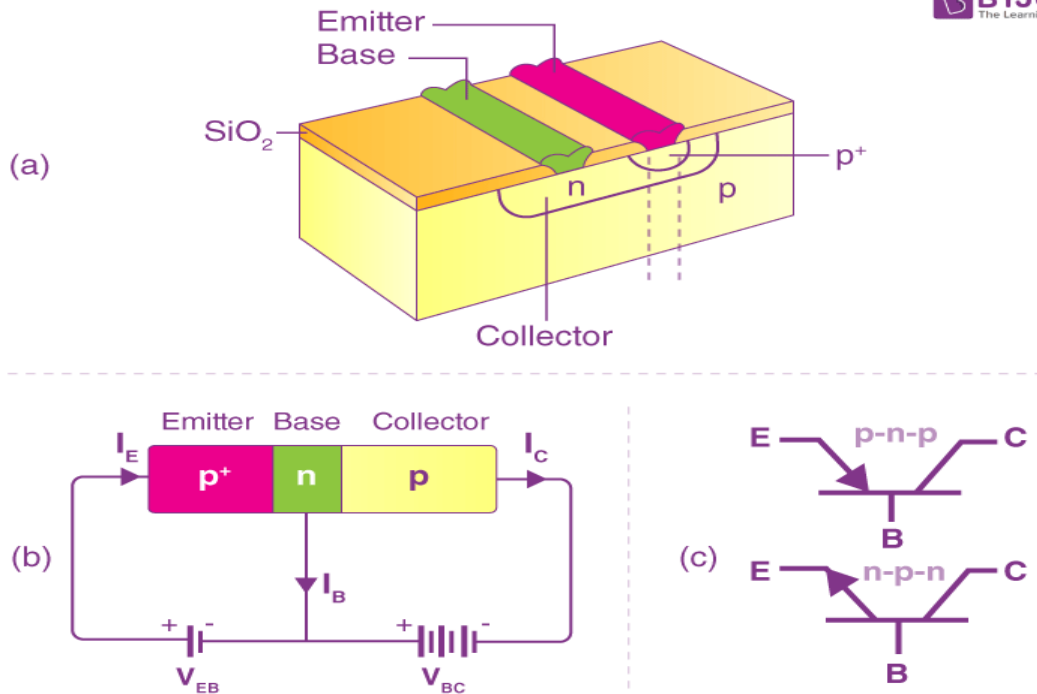
A Bipolar Junction Transistor (BJT) is a semiconductor device used to amplify or switch electronic signals. When designed for operation at microwave frequencies, BJTs are optimized to handle high-speed and high-frequency signals, making them essential in radar, satellite, and communication systems. This report aims to explore the construction, characteristics, and applications of Microwave BJTs.

RF bipolar junction transistors (BJTs) are usually made using silicon (Si), and this transistor is one of the oldest and most popular active RF devices in use today because of its low cost and good operating performance in terms of frequency range, power capacity, and noise characteristics. Silicon junction transistors are useful for amplifiers up to the range of 2–10 GHz, and in oscillators up to about 20 GHz. Bipolar transistors typically have very low $1/f$ -noise characteristics, making them well suited for oscillators with low-phase noise

2. Theoretical Background

A BJT is composed of three regions: emitter, base, and collector. There are two main types: NPN and PNP, with the NPN type being more commonly used in high-frequency applications.

Bipolar junction transistors are sometimes preferred over FETs at frequencies below about 2–4 GHz because of higher gain and lower cost, and the possibility of biasing with a single power supply. Bipolar transistors are subject to shot noise as well as thermal noise effects, so their noise figure is not as good as that of FETs



3. Working Principle:

The operation is based on the movement of electrons and holes. In an NPN transistor, the emitter injects electrons into the base, which are then collected by the collector. The base-emitter junction is forward biased, and the base-collector junction is reverse biased.

ACTIVE DEVICE Is the one require external power source to work.

An active device can do the following:

1. Control current flow
2. Amplify signals
3. Generate oscillations

BJTs as active device in microwaves they amplify and control signal power. Meaning that BJTs takes a small signal at the base and create a larger version of that signal to the collector.

In Microwaves, A BJT (BIPOLAR JUNCTION TRANSISTOR) operates based on controlling a large current flow (collector current) with a smaller current (base current)

A BJT regardless of the operating frequency is a current controlled device

A small base current controls a larger current flow between the emitter and collector.

This control is achieved by the forward bias of the emitter-base (BE) junction and the reverse bias of the collector-base (CB) junction

4.MODES OF MICROWAVE BJT MODES

1. CUT-OFF MODE (OFF STATE)

Base- emitter junction=REVERSE –BIASED

Base-collector junction= reverse biased

No current flow

2. ACTIVE MODE (AMPLIFIER MODE)

Base-emitter = forward biased

Base-collector junction =REVERSE BIASED

BJT transistor amplifies current

3. SATURATION MODE.(FULL ON)

Base-Emitter =FORWARD BIASED

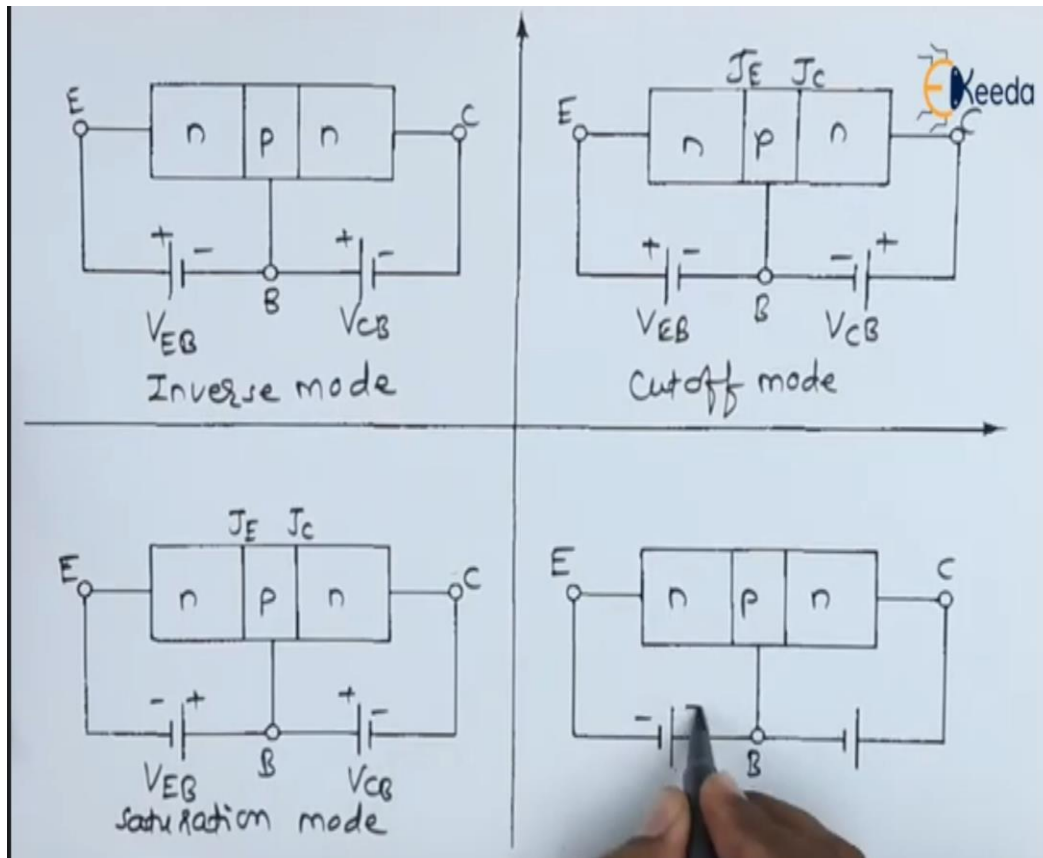
Base-collector junction = forward biased

BJT conducts maximum current.

4. INVERSE ACTIVE MODE (rarely used)

B-E Junction = revers biased

B-C = forward biased



5. Microwave Characteristics of BJT

At microwave frequencies, the performance of BJTs is determined by several factors such as:

1. Gain
2. Noise Figure (NF)
3. Scattering Parameters (S-Parameters)

6.BJT's IN MICROWAVE ENGINEERING

In microwave engineering, BJT is used in:

- Amplifiers (to boost signals)
- Oscillators (to generate signals)
- Mixers (to combine frequencies)

Biasing techniques:

BJTs require proper DC biasing to operate in the active region. Common methods include:

1. Voltage divider bias
2. Collector to base feedback bias
3. Dual power biasing

Impedance matching:

To maximize power transfer and reduce reflections, input and output matching networks are designed based on the S-parameters of the transistor.

7.ADVANTAGES OF BJT IN MICROWAVES

- Can amplify very high-frequency signals
- Easy to model using S-parameters
- Reliable and compact design for RF circuits
- Fast switching due to small base

8.APPLICATIONS OF BJTs

1. SIGNAL AMPLIFICATION

BJTs amplify weak signals by controlling a large collector current with a small base current. This makes them ideal for audio amplifiers, RF amplifiers, and sensor signal boosters.

2. OSCILLATOR CIRCUITS

BJTs are used in oscillator designs (like Colpitts or Hartley oscillators) to produce periodic signals. The transistor provides the necessary gain to sustain oscillations. Common in radio transmitters, RF signal generators, clocks.

3. HIGH-FREQUENCY APPLICATIONS (MICROWAVE AMPLIFIERS)

At microwave frequencies, BJTs (specially designed microwave BJTs like HBTs) are used for low-noise amplification in radar, satellite, and wireless systems.

4. POWER AMPLIFIERS

BJTs are used in RF and audio power amplifiers to drive loads like antennas or speakers. They handle higher current and power levels.

Found in radio transmitters, audio systems, PA systems.

5. ANALOG CIRCUITS (E.G., VOLTAGE REGULATORS)

BJTs are part of feedback-based voltage regulators (like linear regulators). They help maintain a steady output voltage.

Often used in series with load to drop excess voltage.

6. SWITCHING APPLICATIONS (DIGITAL LOGIC & CONTROL)

BJTs can act as electronic switches in logic gates and microcontroller-controlled circuits. In this case, they operate in cutoff (OFF) and saturation (ON) regions.

Used in logic ICs, motor control, LED drivers, etc.

9. challenges and limitations

1. Thermal management
High frequency operations generate significant heats.
2. Frequency limitations
BJTs are gradually replaced by FETS and HBTs
3. Aging effects
Performance degrades due to electromigration and heat stress.

10. conclusion.

Microwave BJTs remain essential components in high-frequency circuit design due to their high gain and efficient performance in amplifier and oscillator applications. Their role in microwave systems continues to be relevant although newer technologies like HBTs and GaN devices are gaining traction.

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